Simple Rules, Complex Behavior KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 2 30 a pattern that repeats chaos emergence the wave triangles

You can use a word more than once.

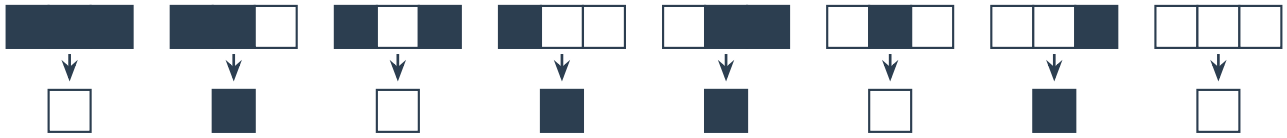
1. The wave. Was anyone in charge of the wave? no

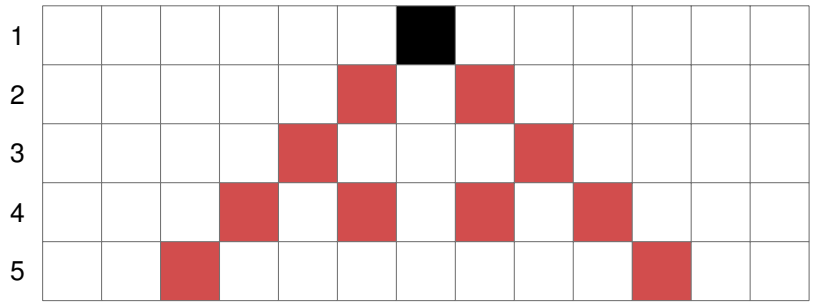
Did the person you watched move down the row? no So what moved? the wave

When many simple parts make something bigger that nobody planned, it's called emergence.

2. Make a few rows. Look at a square and its two neighbors in the row above. Find the same 3 colors on the card. Color the square below. (Outside the grid, everything is white.)

Rule 90: what the middle cell becomes next





What shape is starting to appear? triangles

3. One small change. How many answers are different on the Rule 30 card? 2

Rule 90 makes a pattern that repeats. Rule 30 makes chaos.

The cone snail's shell looks like Rule 30.

Word bank: 2 or 3 3 backward the pattern

4. Explore the rules. Draw each rule. Circle what it looks like.

Rule	What does it look like?			
110	never changes	a pattern that repeats	chaos	complex (complex)
4	never changes	a pattern that repeats	chaos	complex (never changes)
150	never changes	a pattern that repeats	chaos	complex (repeats)
30	never changes	a pattern that repeats	chaos	complex (chaos)

My own rule. My rule number is Answers vary. It looks like _____

Traffic (Rule 184). Every car moves forward. Which way do the jams move? backward

5. Simple rules everywhere.

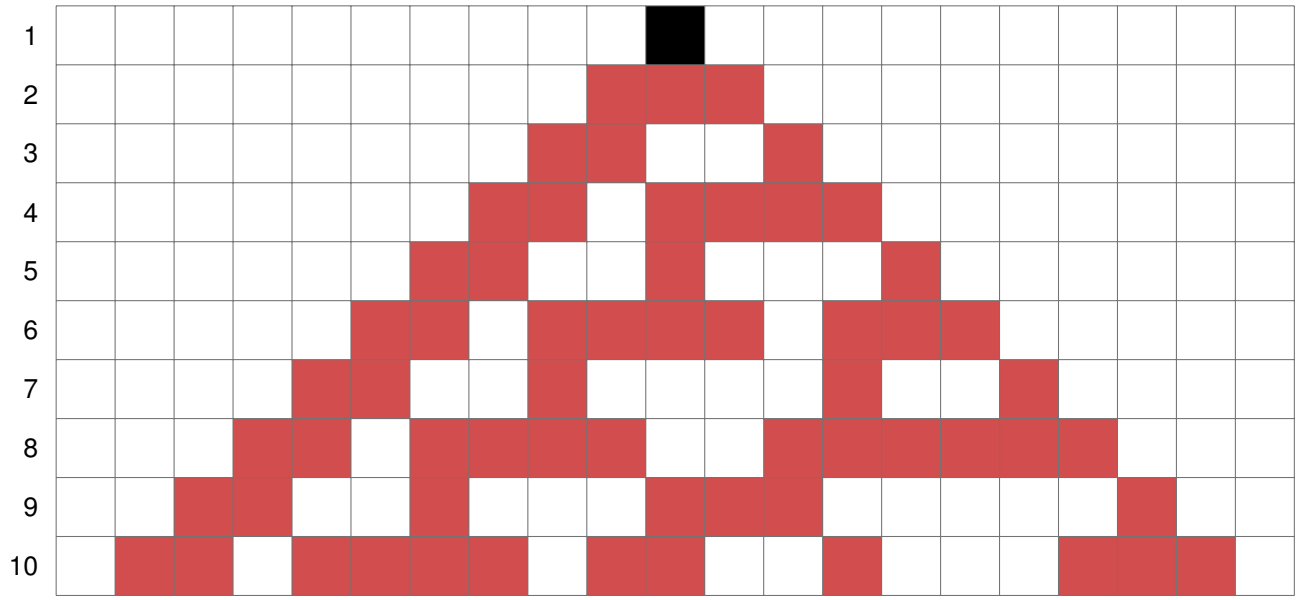
Without "fly the same way," the birds stay in a crowd, but each points its own way.

In the Game of Life, did any single cell move? no So what moved? the pattern

6. Talk about it. Can simple parts make something smart?

I think simple rules **can** / **cannot** make something smart, because Answers vary.

Go Further (Day 2). Be Rule 30 as a class. Color each new row the class shows.



Game of Life. A black cell with 2 or 3 black neighbors stays black. A white cell with exactly 3 black neighbors turns black.